

Enantioselective Access to β -Amino acids via selective C–H bond amination

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Abstract

The incorporation of unnatural β -amino acids as building blocks into pharmaceuticals, including peptides, has shown to improve pharmacokinetic properties, like proteolytic stability. Unlike α -amino acids, β -amino acids are hardly found in nature and thus developing synthetic methods to access these building blocks efficiently and stereoselectively is crucial. Chemical synthesis of β -amino acids often involves laborious procedures, the use of highly functionalized building blocks and wasteful protecting group strategies. In contrast, synthetic strategies based on an enzymatic approach are advantageous due to the exquisite selectivity and efficiency often encountered in enzyme-catalyzed reactions. We aim to develop a heme enzyme that synthesizes β -amino acids through selective C-H bond amination using directed evolution techniques. Enzyme classes, including P411 and *Aeropyrum pernix* protoglobin, have been discovered to introduce amino groups to certain C-H bonds in benzylic and allylic positions. Through additional rounds of directed evolution employing error-prone PCR and site saturation mutagenesis, we aimed to enhance the activity and selectivity of these heme enzymes. Initial activity was showcased using hydroxylamine as the aminating reagents for the enzymatic amination of 3-phenylpropanoic acid sodium salt. Future work will optimize reaction conditions and further evolve the enzyme to improve activity and enantioselectivity.

Introduction

The development of effective medicines across the world heavily relies on synthetic organic chemistry.¹ In this context, biocatalytic approaches have recently emerged as powerful strategies as these operate under mild, environmentally friendly conditions, and utilize nontoxic and biodegradable catalysts while facilitating highly selective chemical transformations.²⁻⁴

Amino acids are the building blocks of proteins and are vital to all organisms. There are 20 naturally occurring amino acids essential for protein synthesis and sometimes energy provision in the human body.⁵ The structural composition of α -amino acids includes a central carbon atom bonded to a carboxyl group, an amino group, a hydrogen atom, and an R group (an organic residue). Due to their importance as fundamental building block of modern life, synthetic access and modification of amino acid derived therapeutical targets have been of longstanding interest. By synthetically altering the core structure of these amino acids, scientists have been able to develop safer pharmaceutical products and implement innovative drug delivery systems.⁶

Amino acids can be categorized based on the number of carbon atoms between the carboxyl and amino group. In α -amino acids, the amino group is attached to the same carbon atom as the carboxyl group. In comparison, β -amino acids have a two-carbon chain between these two functional groups. Although β -amino acids are less common, they hold significant importance in pharmaceuticals due to their stability and resistance to proteolytic enzymes.^{7,8}

Additionally, amino acids can be distinguished by their enantiomeric forms.⁹ This refers to stereoisomers that are mirror images, which are not superimposable, and hence differ in their three-dimensional arrangement. Naturally, amino acids exist in two enantiomeric forms: L and D, classified based on their orientation.¹⁰ L-amino acids are prevalent in the human body. On the other hand, D-amino acids are often found in pharmaceuticals due to their increased resistance to enzyme degradation.^{11,12}

Enzymes are a subclass of proteins with an associated catalytic function.¹³ Enzymes are highly selective as they can often differentiate between enantiomers. In medical applications, pharmaceutical targets which exhibit stereogenic centers are usually used in the enantiopure form, meaning that only one specific stereoisomer is used for treatment. Therefore, it is important to synthesize and modify amino acids or other pharmaceutical molecules with high specificity and stereoselectivity.¹⁴ Currently, the synthesis of β -amino acids is achieved through biocatalytic

reductive amination, as well as other chemical strategies, including the hydrogenation of enamines, the addition of nucleophiles to imines, and the conjugated addition of α,β -unsaturated esters.^{15–27}

The Arnold group has focused their research on finding enzymes capable of catalyzing new-to-nature reactivities and improving their function to provide alternative, more sustainable approaches for the synthesis of relevant organic products. More specifically, the lab uses a diverse variety of heme-binding proteins, commonly found in native hydroxylation and oxidation reactions, to repurpose them in carbene and nitrene transfer reactions. Nitrene transfer reactions enable the addition of amine-containing functional handles onto selected organic targets. The lab has used cytochrome P450-derived variants as a starting point to develop a new class of enzymes, categorized as P411, to facilitate a diverse array of nitrene transfer processes. In this context, the lab has already achieved successful aminations of benzylic positions by various biocatalysts. In 2020, it was found that cytochrome P411 enzyme could introduce amino groups to certain C-H bonds in benzylic and allylic positions.^{28,29} In later studies, a different class of heme enzymes, *Aeropyrum pernix* protoglobin (*ApePgb*), have been found to perform similar non-native reactions, such as carbene transfer.³⁰ This enzyme class was recently identified to also enable the amination of benzylic C-H bonds to synthesize β -amino acids.

Currently, we are seeking enzyme variants that are capable of selectively introducing primary amines at the β -position of ester substrates to access the corresponding β -amino acids building blocks. Especially variants that exhibit high regioselectivity and enantioselectivity are desirable to eventually facilitate the formation of both enantiopure amination products. The goal is to develop a biocatalytic methodology to functionalize unreactive C-H bonds. After testing various substrates and optimizing reaction conditions, we successfully found an enzyme variant derived from *Aeropyrum pernix* protoglobins that regioselectively functionalized sodium 3-phenyl propanoate to generate the corresponding β -amination product. As an aminating reagents, hydroxylamine hydrochloride could be utilized.

Results and Discussion

First, we performed one round of directed evolution with the parent variant G3, determined from previous rounds of evolutions. In this process, we introduced mutations through error prone polymerase chain reaction (epPCR), followed by gel electrophoresis to purify the DNA constructs. Then, Gibson assembly was used to generate a circular plasmid by inserting the diversified gene product into the pET22b(+) vector with ampicillin resistance. Transformation with *E. coli* BL21 were performed, and single colonies were picked into 96-well plates. Overexpression of the recombinant variants enabled us to harvest these new protein variants. We then performed enzymatic reactions and analyzed the crude reaction mixture via HPLC analysis. By HPLC-MS analysis, both the α - and β -amination product were identified by mass spectrometry and could be separated as they exhibit different retention times based on their different physical properties. To validate the results that were obtained from the 96-well plate screen, the most promising variants were reevaluated under normalized reaction conditions (OD_{600} was set to 30 for performing the reaction). This comprehensive analysis provided insights into product formation and the types of products formed for selected variants, as illustrated in Figure 1.

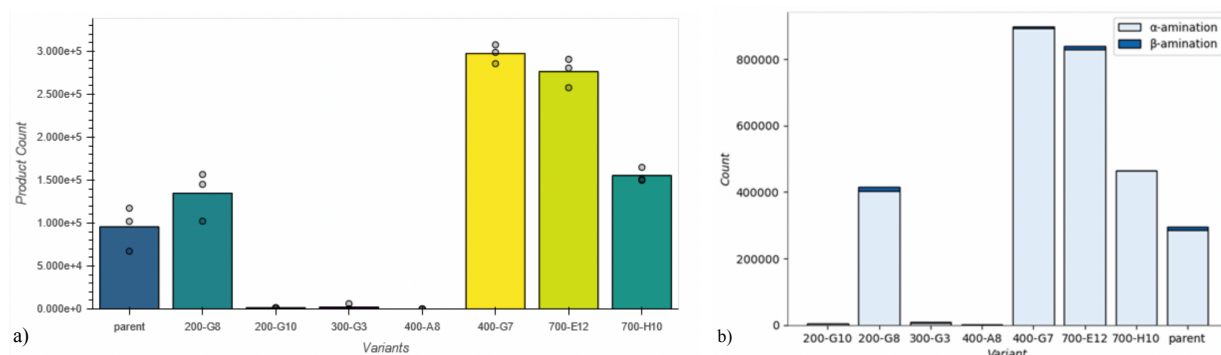
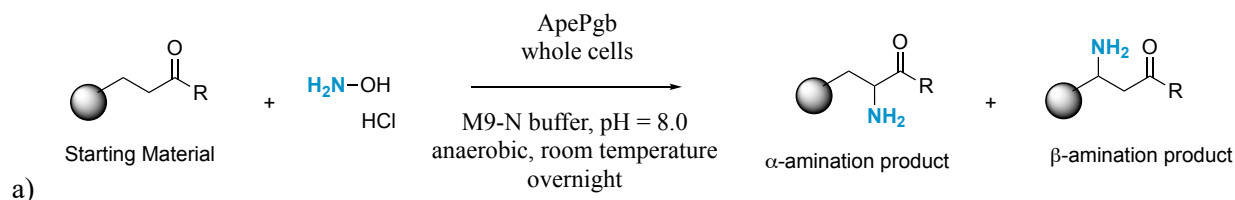


Figure 1. a) Initial activity of selected variants obtained from first round of epPCR. b) Analysis of α -amination and β -amination activity across selected variants obtained from first round of epPCR.

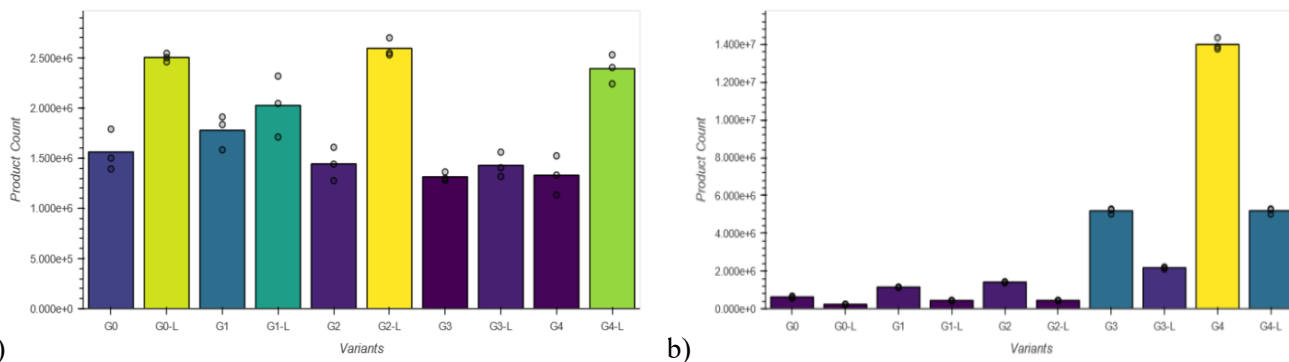
These results show new enzyme variants with regioselectivity for the α -amination product. Although variant 400-G7, mutated under 400 μ M of Manganese Chloride (MnCl_2), showcased the highest activity, it did not showcase the desired regioselectivity. Looking for different activity we decided to test a variety of substrates regarding their promiscuity toward amination that led to either the α - or β -amination products. Relying on variants that have shown improved activity towards the benzylic amination of ester substrates and using hydroxylamine as the aminating reagent; four different substrates were submitted to both whole cell and lysate reactions harboring the *ApePgb* variants. The general structure of the reactions is shown in Figure 2a, while the four different substrates used are shown in Figure 2b. To set up these reactions, all variants, in their whole cell and lysate form, were distributed over two 96-well plates. Then, 20 μ L of aqueous glucose stock was added to each well. The substrates (20 μ L each, 100 mM stock in ethanol) and the hydroxylamine hydrochloride (20 μ L, 200 mM stock in water) were added inside the coy chamber since these reactions are performed under anaerobic environment. Analysis of the reaction relied on the use of HPLC-MS through mass spectrometry. This comprehensive analysis provided insights into product formation and distributions (α - vs. β -amination product) for selected variants. Figure 3 shows the activity levels of product formation for the whole variant lineage under starting material of the sodium 3-phenyl propanoate (sodium salt of the carboxylic acid), a) shows the α -amination product while b) shows the β -amination product.



b)

Starting Material	β -amination Product	Product Detected
 Phenyl ester		Yes
 Carboxylic Acid Sodium Salt		Yes
 Alkene ester		No
 Alkenyl ester		No

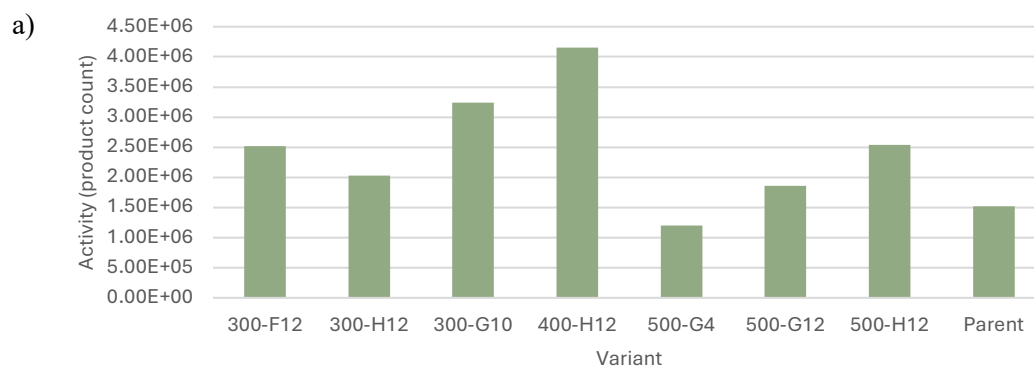
Figure 2. a) Model reaction for all four validation reactions. Colored red changes between reactions. b) Table showing all substrates used as well as the desired β -amination product and whether the corresponding product mass was detected.



a) Analysis towards the formation of the α -amination product of whole cell and lysate reactions of the carboxylic acid sodium salt substrate. b) Analysis towards the formation of β -amination product of whole cell and lysate reactions of the carboxylic acid sodium salt substrate.

Overall, the results for most substrates showcased higher activity under whole cell reaction conditions when compared to lysate reaction conditions. This was not expected since the cell membrane usually acts as a barrier and diffusion of reagents is frequently more challenging. However, since whole cell showed the more promising results, we chose that type of reaction to run for the next round of directed evolution. Figure 3 showcases enzyme variants labeled G3 and G4 as the most selective towards the β -amination product.

Following these results, we continued to perform another round of directed evolution using the sodium salt of the carboxylic acid, the substrate that showed the better results in the previous validation experiment. Using variant G3--the variant with the high selectivity--as parent, we introduced mutations through epPCR, and continued with the process of directed evolution, and later analyzed the α - and β -amination product by HPLC-MS analysis. Figure 4 shows the results of the enzyme variants ran for validation, where a) shows the product formation while b) shows the present mutations for selected variants.



b)

Variant	Mutations
Parent (G3)	-
300-1-F12	Parent
300-1-H12	P188R
300-2-G10	G117E
400-1-H12	E15K, E67G
500-1-G4	T30A, R122C
500-1-G12	V47A, K49E, C102S, W109*, E159G, S163P, E168G
500-2-H12	N75S, E80G, T137N, R140H, D194V, W195*

Figure 4. a) Product formation of selected variants obtained from directed evolution. b) Mutations present in each variant when compared to parent DNA

Figure 4 showcases improved activity of variant H12, sourced from a MnCl_2 concentration of 400 μM epPCR plate. When comparing these results to the mutations observed, this variant was different from the parent variant by only one amino acid. Additionally, from the other mutations results, there are two variants that experienced a truncating mutation, where a stop codon was introduced into the DNA sequence. This behavior was found for variants G12 and H12 both generated at 500 μM of MnCl_2 .

Taking these results into account, we used 400-H12 enzyme variant to parent another round of directed evolution using site-saturation mutagenesis (SSM), a strategy to introduce mutation in a focused manner. This strategy permits purposeful mutation of specific sites in the amino acid sequence. Sites 93 and 145 are located at the active site of the enzyme, making them desirable sites to selectively mutate. Using these sites, the same protocol as epPCR was followed with a variation on the PCR set-up. Figure 5 shows the results obtained from this round of directed evolution.

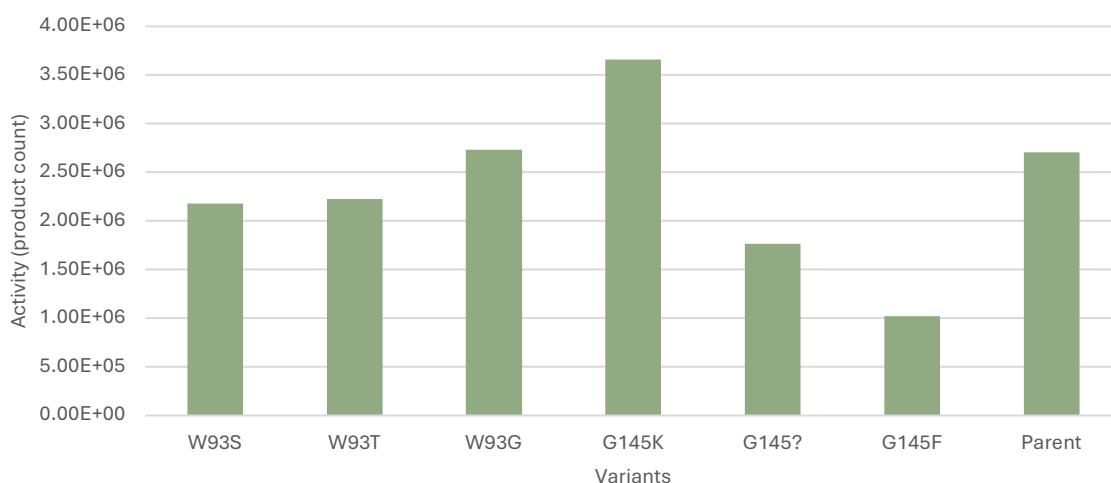
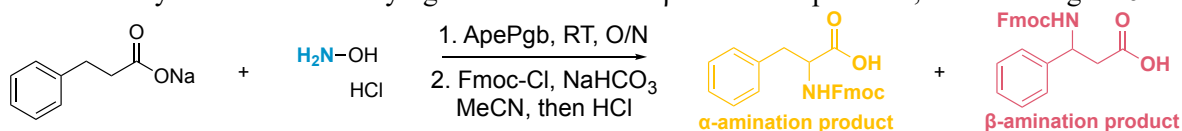


Figure 5. Product formation of selected variants obtained from directed evolution.

From Figure 5, G145K is the most active variant, and to further analyze the specificity of the enzyme, 9-fluorenylmethyloxycarbonyl (Fmoc) group was used to protect amino group in the product, as shown in Figure 6a, and HPLC-MS analysis aided in identifying each of the α - and β -amination products, shown in Figure 6b.



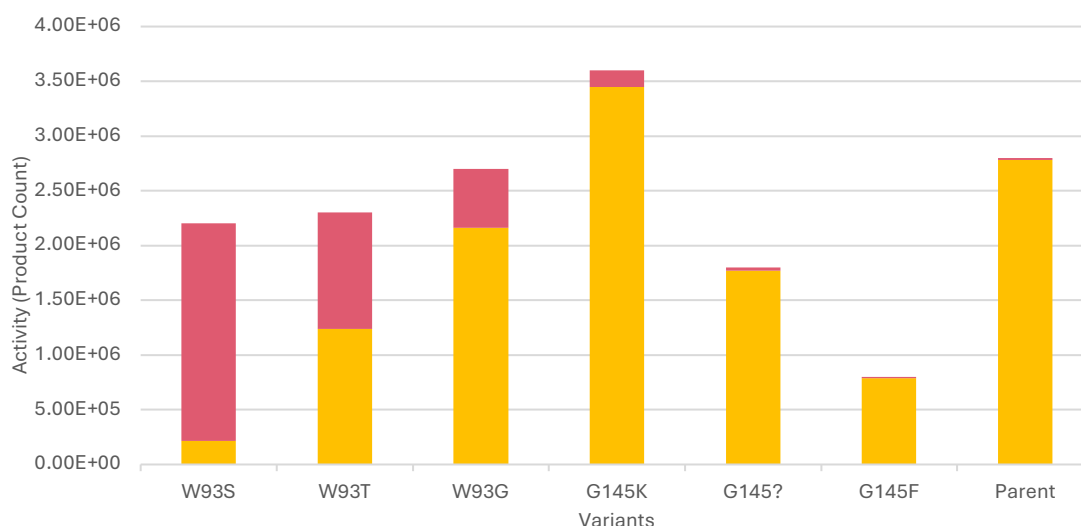


Figure 6. a) Model reaction for separation for products for regioselective analysis. b) Analysis of α -amination (yellow) and β -amination (red) activity across selected variants obtained from the SSM round of directed evolution.

The results shown in Figure 6b show that although variant G145K was overall more active, it was not selective towards the β -amination product, whereas variant W93S, mutated at site 93, showcased high levels of regioselectivity towards the desired product. We successfully evolved regioselective variants, at the α -position and β -position, with hydroxylamine as the aminating reagent. The next steps would lead to an exploration of variant W93S and its evolution toward higher activity and selectivity, including enantioselectivity.

Methods

To improve the activity and selectivity of an initially active enzyme variant, we performed directed evolution (Figure 7). Directed evolution is a method that mimics the natural evolution of enzymes in combination with selection in a more controlled and directed environment. This method helps evolve proteins towards a defined goal. One of the key aspects in directed evolution is the process of gene diversification to access novel protein variants. In general, two different methods are used to generate evolved variants from initial active enzymes: error prone polymerase chain reaction (epPCR) and site saturation mutagenesis (SSM).

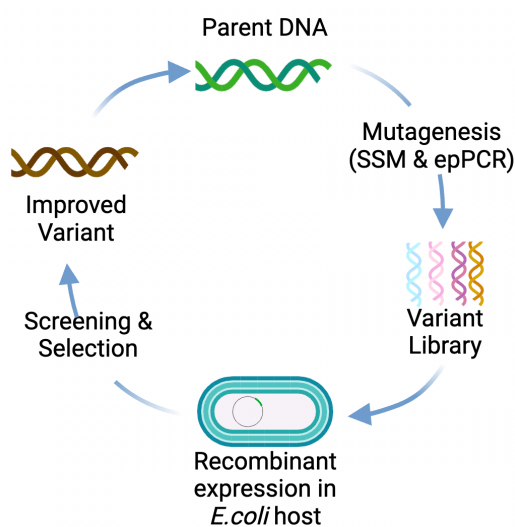


Figure 7. Directed evolution of enzyme variants.

Method 1: In epPCR experiments, random point mutations are introduced along the gene of interest while being amplified, using a low fidelity polymerase. Different PCRs, with an elongation time of 1.30 min/kb, are run each under the same conditions except for the concentration of manganese chloride (MnCl_2), which ranged from 200mM to 500mM to reduce the efficiency of the Taq polymerase.

Method 2: SSM is a strategy to introduce mutations in a focused manner, thus enabling us to purposefully replace a single amino acid at a specific site in the amino acid sequence. SSM relies on randomizing at a chosen position on the DNA fragment. Two overlapping primers are produced to make one mutated segment through Gibson assembly.

Irrespective of which type of mutagenesis, the PCR is worked up with DpnI digest to remove parent DNA and later purified through gel purification. To assemble the circular plasmids, the Gibson Assembly Master Mix (1.33x) is used alongside the purified DNA (two segments for SSM) and the backbone (pET22b(+)) vector for epPCR. Transformation is done with chemically competent cells, *E. coli* BL21, and single colonies are later picked into 96 deep well plates with 400 μL of Luria Broth (LB) media and ampicillin. They were incubated overnight at 37 °C and 220 rpm. To prepare preexpression cultures, 50 μL from the overnight culture are transferred into another set of 96 deep well plates with Terrific Broth (TB) media with ampicillin. The cells are incubated for two hours at 37 °C and 220 rpm.

Protein expression is induced by the addition of 50 μL of α -liponic acid (ALA, 1000x stock) and 50 μL of isopropyl β -D-1-thiogalactopyranoside (IPTG, 1000x stock). The cultures are then again incubated for 16-22 hours at 22°C and 220 rpm.

To set up the reaction, the overnight cultures are taken and centrifuged at 4000 rpm for 10 min at 4 °C to harvest the proteins. The supernatant is removed and the pellets usually appeared in different shades of pink, confirming the expression of the heme-containing protoglobins. The pellets are then resuspended in 360 μL of M9-N buffer, a minimal salts buffer at pH 8 with 0.002 mg of glucose. The reaction is run overnight in the Coy chamber after the addition of 5 mM substrate (100 mM stock solution, 20 μL) and 10 mM of the aminating reagent (200 mM stock solution, 20 μL).

The reaction is worked up by being quenched with 800 μL of acetonitrile (MeCN), after the plates are then centrifuged to pellet the cell debris, and only 200 μL of the supernatant is transferred to an HPLC plate for analysis. Analysis of the reaction relied on the use of HPLC-MS through mass spectrometry. This comprehensive analysis provided insights into product formation and distributions (α - vs. β -amination product) for selected variants.

The results help determine which variants are the most effective, and after selecting a more targeted group of variants, the experiment is performed once more to validate the results under more controlled settings. The key changes to experimental design while validating are larger volumes and more exact measurements of OD.

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